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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Time: 03 Hours

Code: 100111 ***Subject: Programming for Problem Solving*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) Which of the following is a valid variable name in C?
- (i) 2value
- (ii) @value
- (iii) value_2
- (iv) value#2

(b) Which of these is not a valid data type in C?
- (i) int
- (ii) real
- (iii) float
- (iv) char

(c) Which of the following is the correct syntax for if statement?
- (i) if [condition] then
- (ii) if condition then
- (iii) if (condition)
- (iv) if {condition}

(d) What is the output of printf("%d", 5 + 3 * 2)?
- (i) 16
- (ii) 11
- (iii) 10
- (iv) 13

(e) Which header file is required to use printf()?
- (i) stdlib.h
- (ii) conio.h
- (iii) stdio.h
- (iv) string.h

(f) Which keyword is used to define a constant in C?
- (i) final
- (ii) constant
- (iii) #define
- (iv) static

(g) What is the index of the first element in an array?
- (i) 1
- (ii) -1
- (iii) 0
- (iv) Depends on array

(h) Which one is the logical AND operator in C?
- (i) &
- (ii) |
- (iii) &&
- (iv) ||

(i) Which format specifier is used for printing float values?
- (i) %d
- (ii) %c
- (iii) %f
- (iv) %x

(j) Which of these is not a loop in C?
- (i) for
- (ii) while
- (iii) do-while
- (iv) repeat-until

Q.2 Solve both questions : **[7+7=14]**

(a) Explain the structure of a C program with an example.

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(b) Describe basic data types in C with memory size.

Q.3 Solve both questions : **[7+7=14]**

(a) What is an algorithm. Write an algorithm to perform the addition of two numbers in C and draw a flowchart to represent the addition of two numbers.

(b) Write a program to check whether a number is even or odd using if-else.

Q.4 Solve both questions : **[7+7=14]**

(a) Explain Arithmetic and Logical operators in C with example programs.

(b) Explain Relational and Bitwise operators with example programs.

Q.5 Solve both questions : **[7+7=14]**

(a) Explain call by value and call by reference with examples.

(b) What is the difference between library functions and user-defined functions?

Q.6 Solve both questions : **[7+7=14]**

(a) What is an array in C programming? Write advantages of using arrays. Explain a one-dimensional array with a suitable example program.

(b) Describe the difference between character array and string.

Q.7 Solve both questions : **[7+7=14]**

(a) Write the difference Between Array and Structure in C programming.

(b) Write the difference Between Structure and Union in C programming.

Q.8 Solve both questions : **[7+7=14]**

(a) What is a pointer? Explain pointer declaration and initialization with sample program.

(b) What is file? Explain text file and binary file in details.

Q.9 Write short notes on any two of the following: **[7x2=14]**

(a) Structure
(b) Union
(c) Recursion
(d) Fibonacci series program

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